DGMS 2019 • SECTION H: ICHNOLOGY, REMOTE SENSING & GIS

Question 66: Seilacherian Ichnofacies & The Bathyal Zone Zoophycos Assemblage

Marine Bathymetric Depth Zonation, Trace Fossil Ethology, and Sedimentary Substrate Controls

OFFICIAL EXAMINATION QUESTION

Zoophycos assemblage is the characteristic ichnofacies of which one of the following zones?

(a) Sublittoral zone

(b) **Bathyal zone [CORRECT]**

(c) Abyssal zone

(d) Littoral zone

VERIFIED ANSWER KEY

Correct Option: (b) Bathyal zone — In the classical marine bathymetric ichnofacies model established by Adolf Seilacher (1967), the **Zoophycos ichnofacies** is characteristic of low-energy, quiet-water continental slope settings corresponding to the **bathyal depth zone** (typically 200 m to 2000 m depth, below storm wave base), situated between the shallow shelf (Cruziana ichnofacies) and deep ocean abyss (Nereites ichnofacies).

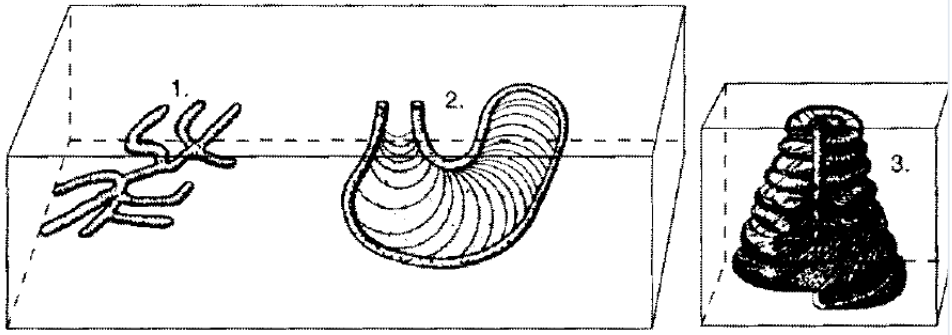


Figure 4.4.3
Trace fossil association characteristic of the *Zoophycos* ichnofacies. 1) *Phycosiphon*, 2
Zoophycos, 3) *Spirophyton*. [From Pemberton, S. G., J. A. MacEachern, and R. W. Fre
1992, Trace fossil facies models: Environmental and allostratigraphic significance, *in*
cies models, Walker, R. G., and N. P. James (eds.), Geological Association of Canada
Fig. 11, p. 55. Reproduced by permission.]

NEREITES ICHNOFACIES

The *Nereites* ichnofacies is characteristic of deep water and is apparently
stricted to turbidite deposits. It is distinguished by complex horizontal cra
ing and grazing traces and patterned feeding or dwelling structures.
ichnogenera are ornate and complicated, such as *Paleodictyon*, *Spirorhaphe*, a
Nereites (Fig. 4.4.4). Total diversity of traces is high, but the abundance of in
vidual traces is low. The Nereites ichnofacies develops initially in sandy (t
bidite) substrates but may later colonize parts of some muddy (pela
deposits that form on top of sandy turbidites.

OTHER ICHNOFACIES

The *Psilonichnus* ichnofacies (Fig. 4.4.5) is a softground ichnofacies develo
under nonmarine to very shallow marine or quasi-marine conditions. I
characterized by J-, Y-, or U-shaped burrows of marine organisms, vert

Figure 66.1 | Bathymetric Distribution of Seilacherian Ichnofacies: Schematic bathymetric profile across the continental margin illustrating the succession of marine trace fossil facies: Littoral / foreshore (*Skolithos*), Sublittoral shelf (*Cruziana*), Continental slope / **Bathyal zone** (***Zoophycos***), and Deep marine abyssal plain (*Nereites*). (Source: Sam Boggs Jr., *Principles of Sedimentology and Stratigraphy*, 4th Ed., Fig. 4.38, p. 104; after Ekdale, Bromley & Pemberton, 1984; Crimes, 1975).

1. Principles of Ichnology & Seilacherian Ichnofacies

An **ichnofacies** is an association of trace fossils that recurs through geological time under similar environmental, depositional, and bathymetric conditions. Pioneered by Adolf Seilacher (1967), trace fossil classification reflects organism behavior (ethology) in response to

hydrodynamic energy, sedimentation rate, substrate consistency, food availability, and oxygenation.

The classic depth-related softground marine ichnofacies succession across a continental margin includes:

1. **Psilonichnus Ichnofacies:** Supratidal and backshore dunes, beaches, and washover fans. Characterized by crab burrows (e.g., *Psilonichnus*) and root traces.
2. **Skolithos Ichnofacies (Littoral to shallow sublittoral):** High-energy intertidal to sandy upper shoreface. Dominance of vertical, cylindrical dwelling burrows (*Domichnia*) constructed by suspension feeders (e.g., *Skolithos, Ophiomorpha, Diplocraterion*) to withstand shifting sands and wave agitation.
3. **Cruziana Ichnofacies (Sublittoral zone / Shelf):** Moderate-to-low energy shelf below fair-weather wave base but above storm wave base. Diverse locomotion traces (*Repichnia*), resting traces (*Cubichnia*), and feeding burrows (*Fodinichnia*) of deposit and suspension feeders (e.g., *Cruziana, Asteriacites, Rhizocorallium, Teichichnus*).
4. **Zoophycos Ichnofacies (Bathyal zone / Continental Slope):** Quiet, low-energy continental slope and rise environments with muddy substrates and dysaerobic (low oxygen) pore-water conditions. Complex planar to helical deposit-feeding structures (*Fodinichnia*), predominantly *Zoophycos, Phycosiphon*, and *Spirophyton*.
5. **Nereites Ichnofacies (Abyssal zone / Deep Ocean Plain):** Deep abyssal plains and distal turbidite basins (>2000 m). Dominance of intricate, highly systematic grazing trails (*Pascichnia*) and farming burrows / graphoglyptids (*Agrichnia*) designed to maximize foraging efficiency in food-starved settings (e.g., *Nereites, Paleodictyon, Spirorhaphe, Cosmorhaphe*).

chnofossils, or lebensspuren. Study of trace fossils constitutes the discipline of **logy**, which has become increasingly complex since the mid-1950s and has ıed a massive body of literature. Several of these books are listed under ıer Reading" at the end of this chapter.

of Trace Fossils. Trace fossils are not true bodily preserved fossils; that is, lo not form by conversion of a skeleton into a body fossil. They are simply ures that originated through the activities of organisms. Interpreted broadly, ıic structures can be considered to include the following: (1) bioturbation ures (burrows, tracks, trails, root penetration structures), (2) biostratification ures (algal stromatolites, graded bedding of biogenic origin), (3) bioero- ructures (borings, scrapings, bitings), and (4) excrement (coprolites, such as ellets or fecal castings). Not all geologists regard biostratification structures e fossils, and these structures are not commonly included in published dis- ns of trace fossils.

Trace fossils are classified into **ichnogenera** on the basis of characteristics elate to major behavioral traits of organisms and are given generic names as *Ophiomorpha*. Distinctive but less important characteristics are used to y **ichnospecies**, e.g., *Ophiomorpha nodosa*. Trace fossils are produced by a f marine organisms such as crabs, flatfish, clams, molluscs, worms, shrimp, el. In nonmarine environments, organisms such as insects, spiders, worms, edes, snails, and lizards can produce a variety of burrows and tunnels; ver- e organisms leave tracks; and plants leave root traces. The organisms that ce traces are rarely preserved with the traces; thus, the trace maker is com- not known. Therefore, the names applied to ichnogenera and ichnospecies ılly do not refer to the trace makers themselves.

ossil Assemblages. From a sedimentological standpoint, study of trace-fossil blages has commonly proven to be more useful than study of individual genera or ichnospecies. A trace-fossil assemblage is a basic collective term nbraces all of the trace fossils present within a single unit of rock. Although ıs kinds of trace-fossil assemblages are recognized, grouping of trace fossils **chnofacies** has particular significance in paleoenvironmental studies. ıer (1964, 1967) introduced the concept of ichnofacies to describe associa- of trace fossils that are recurrent in time and space and that reflect environ- l conditions such as water depth (bathymetry), salinity, and the nature of bstrate in or on which they formed (e.g., mud vs. sand bottom). Fundamen- chnofacies are sedimentary facies defined on the basis of trace fossils, and chnofacies may include several ichnogenera.

Figure 66.2 | Comprehensive Ichnofacies Classification: Diagnostic characteristics, ethological groups, representative ichnogenera, and typical depositional environments of softground, firmground, hardground, and woodground trace fossil suites. (Source: Sam Boggs Jr., *Principles of Sedimentology and Stratigraphy*, 4th Ed., Table 4.3, p. 103).

2. Ethology & Environmental Dynamics of *Zoophycos*

Zoophycos consists of an elaborate, three-dimensional spiral or planar spreite structure radiating from a central axis. A worm-like deposit-feeding organism systematically mined nutritious

organic-rich mud below the sediment-water interface, creating a corkscrew-like spreite through repetitive back-and-forth passes.

As documented by **Boggs (2006, Chapter 4, pp. 105–108)**:

- **Bathymetric Locus:** Predominantly continental slope, toe of slope, and outer shelf settings (bathyal depths, ~200–2000 m).
- **Oxygen Conditions:** Commonly associated with oxygen-deficient (dysaerobic) conditions where surface-grazing epifauna cannot survive, leaving deep sediment tiers to specialized *Zoophycos* miners.
- **Substrate:** Fine-grained, cohesive mud and silty hemipelagic softgrounds.

3. Comprehensive Option Analysis

Option	Marine Bathymetric Zone	Characteristic Seilacherian Ichnofacies	Evaluation
(a) Sublittoral zone	Continental shelf (low-tide line to ~200 m depth)	**Cruziana ichnofacies**	*Incorrect.* Characterized by diverse resting (*Asteriacites*) and crawling (*Cruziana*) traces of shelf fauna.
(b) Bathyal zone	Continental slope and rise (~200 m to 2000 m depth)	**Zoophycos ichnofacies**	**Correct.** Classical home of the *Zoophycos* assemblage; low-energy, dysaerobic muddy slope.
(c) Abyssal zone	Deep ocean floor (>2000 m to 6000 m depth)	**Nereites ichnofacies**	*Incorrect.* Dominated by farming networks (*Paleodictyon*) and systematic grazing spirals (*Spirorhaphe*).
(d) Littoral zone	Intertidal / foreshore zone (between high and low tides)	**Skolithos ichnofacies**	*Incorrect.* Dominated by robust vertical tubes (*Skolithos, Diplocraterion*) in wave-swept sands.

4. Authoritative Textbook References

- **Sam Boggs Jr. (2006)** *Principles of Sedimentology and Stratigraphy*, 4th Ed., Pearson / Prentice Hall, Chapter 4 ("Sedimentary Structures"), Section "Biogenic Sedimentary Structures / Ichnofacies", pp. 102–109, Fig. 4.38 & Table 4.3.
- **Adolf Seilacher (1967)** "Bathymetry of trace fossils", *Marine Geology*, 5(5–6): 413–428.
- **Pemberton, MacEachern & Frey (1992)** "Trace fossil facies models: Environmental and allostratigraphic significance", in Walker & James (eds.), *Facies Models: Response to Sea Level Change*, Geological Association of Canada, pp. 47–72.
- **R. G. Bromley (1996)** *Trace Fossils: Biology, Taxonomy and Applications*, 2nd Ed., Chapman & Hall, London, pp. 240–255.